



Opening the Black Box: Mechanistic Interpretability of Agent Tool Selection with Sparse Autoencoders

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Key Result — Causal, Not Just Correlated

10.1 %

of agent tool choices reverse when 10 SAE features are ablated.

Removing a handful of features *causally changes* the agent's tool choice.

random ablation: 0 % · SAE round-trip: 0 % · $p = 0.002$

also significant: single vs. multi-tool — 17.1 % vs. 2.4 % baseline ($n = 41$, $p = 0.029$).

fundamental vs. technical analysis (finance domain, 1 of 23 contrast types tested) — Nemotron-3-Nano-30B, layer 20.

The Tool-Selection Black Box

Modern AI agents autonomously pick among databases, code, web search, and sub-agents. When the agent chooses, *why* it chose remains opaque.

- **Prompt engineering:** correlational, not causal.
- **Behavioural testing:** input/output only.
- **Chain-of-thought:** plausible, often confabulated.

We open the black box at the token where the model commits to a tool and decompose its hidden state into human-readable features.

Pipeline

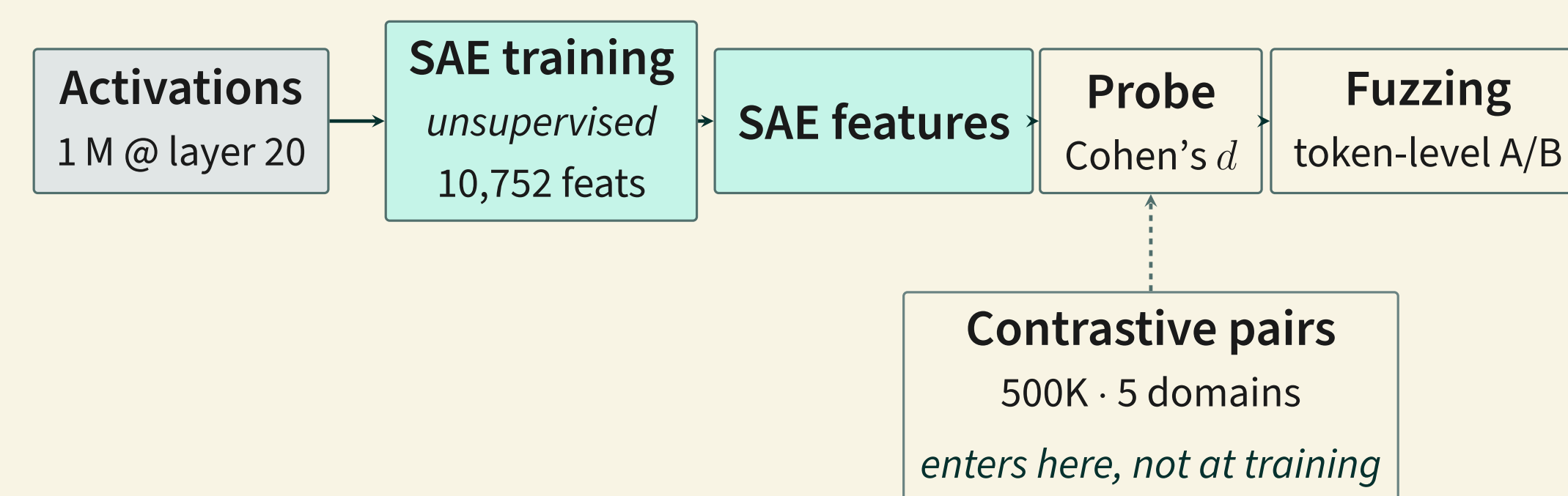


Figure 1. The SAE is trained without contrastive pairs. Pairs enter only at the probe step to surface which pre-existing features carry decision-relevant signal — never as a training signal.

Pick the Right Layer (Not the MoE Ones)

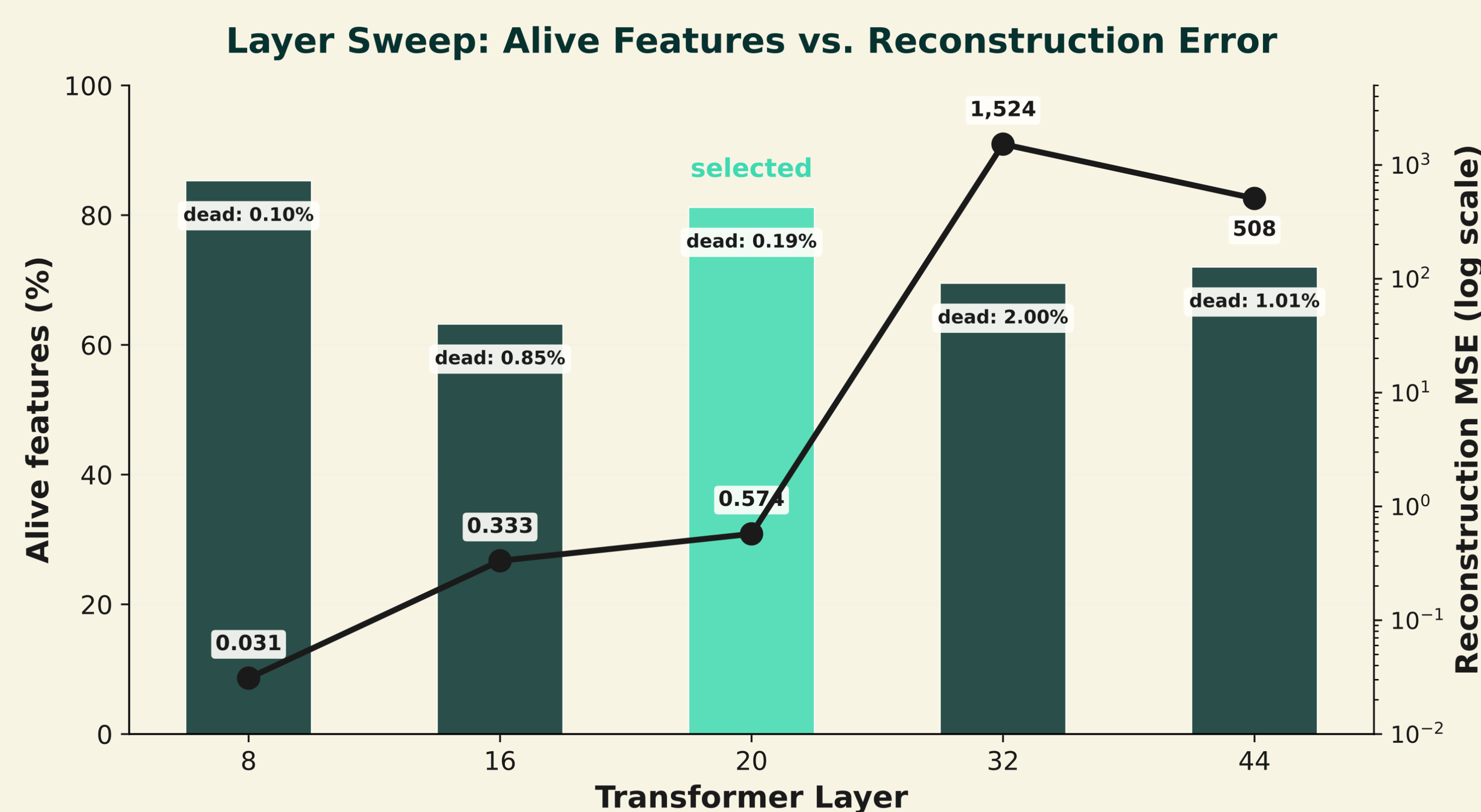


Figure 2. A single SAE cannot decompose MoE layers. Reconstruction MSE explodes by $>500\times$ at layers 32 and 44 (sparse expert routing); layer 20 sits just before the MoE stack and gives 81.2 % alive features. Don't probe MoE layers with one SAE.

JumpReLU Sparse Autoencoder

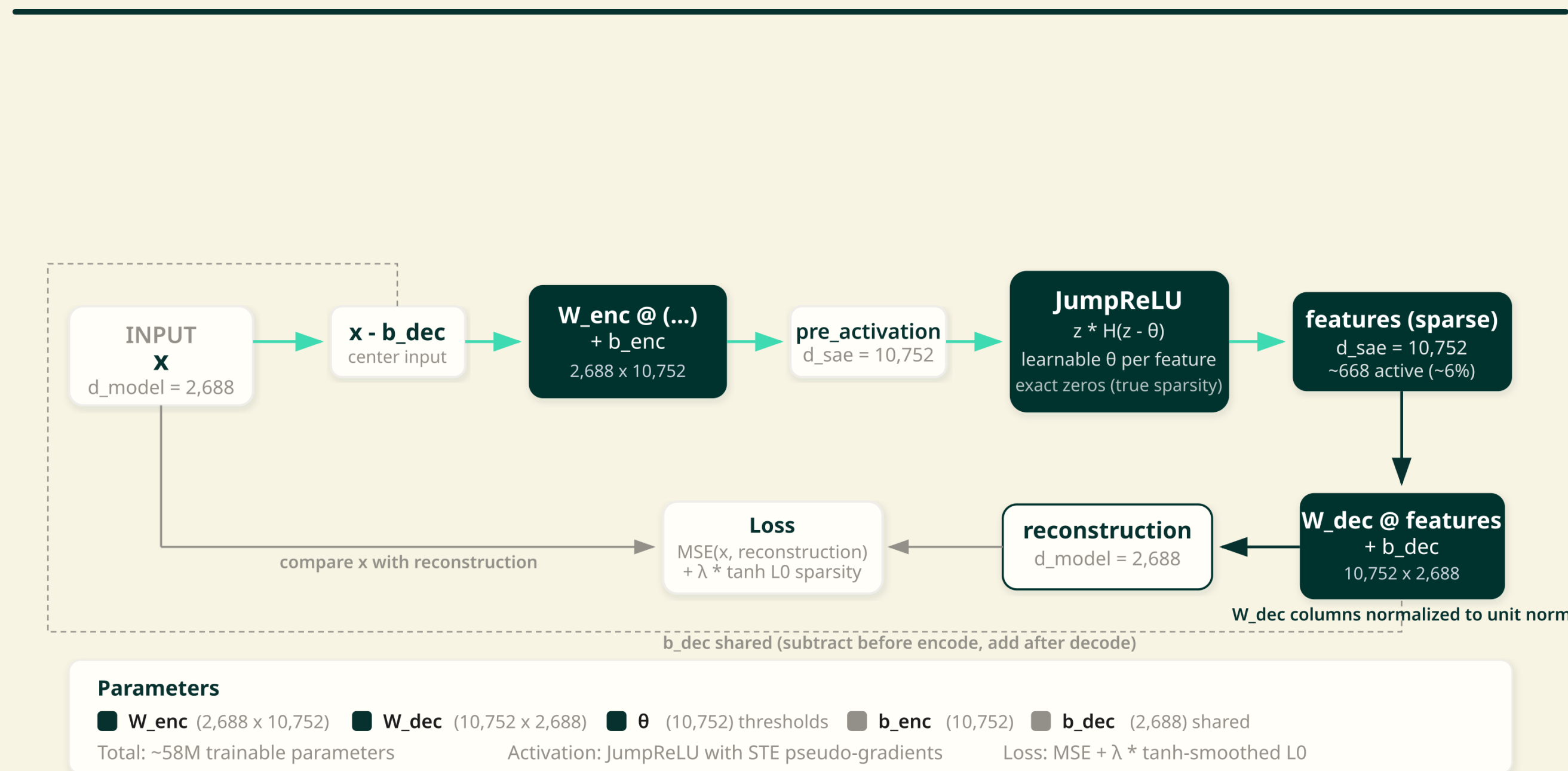


Figure 3. An overcomplete dictionary that decomposes hidden states into monosemantic features. JumpReLU thresholds θ_i produce exact zeros (true sparsity), trained with a smooth tanh surrogate [2].

Dictionary M	10,752 ($4\times$ hidden dim)
Activations trained on	1,000,000
Reconstruction MSE	0.574
Alive features	81.2 %
Active per input	668 ± 26 (6.2 %)

Inside a Feature — Emergent, Unsupervised Concepts

The SAE decomposes Layer-20 activations into 10 752 features. Without contrastive supervision, named concepts emerge from raw activations alone:

- “internal knowledge retrieval”
- “data modification intent”
- “query complexity”

Each feature carries an LLM-generated label, a token-level fuzzing score (aggregate 0.912 ± 0.008 , $p < 10^{-4}$), and its row in the contrastive-ablation table.

Scope & Limitations

- One model (Nemotron-3-Nano-30B), one layer, one decision-token convention.
- Top-10 ablation is conservative; distributed decisions need larger sets.
- The Qwen3-VL-235B pipeline (pair generation, feature labelling, fuzzing judge) is the single source of LLM supervision — model errors propagate to rationales.

Why This Matters in Production

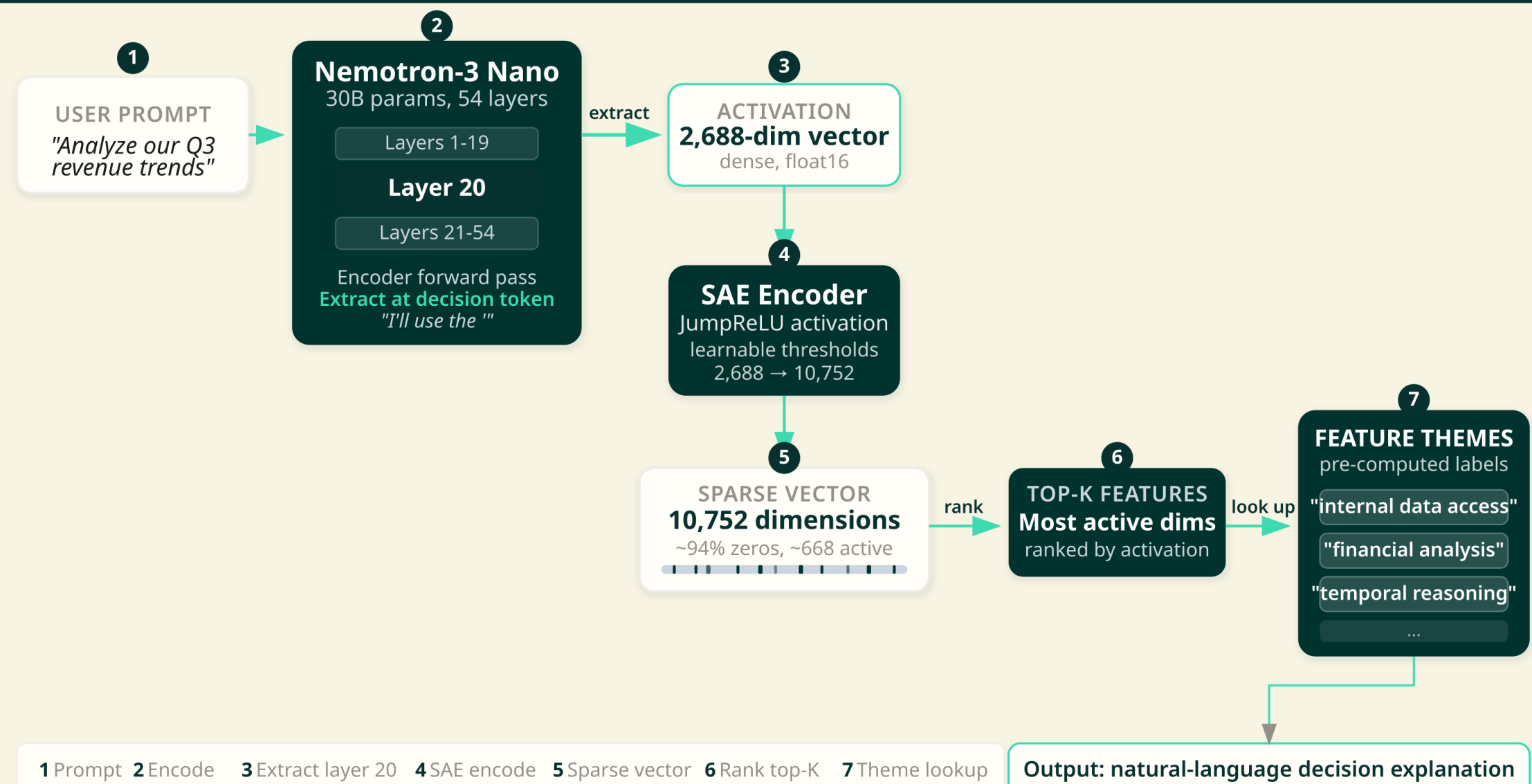


Figure 4. Every tool call ships with a rationale. At inference, the top- K active features at the decision token map to pre-computed labels — a human-readable explanation alongside the tool choice, with no extra LLM call.

Most SAE work stops at finding features. Here the features **ship to inference**: every tool call carries a top- K feature rationale alongside it — a **debugging surface** for deployed agents, and a foundation for **future anomaly detection** when activations deviate from a tool's normal fingerprint.

Feature Label Quality (Token-Level Fuzzing)

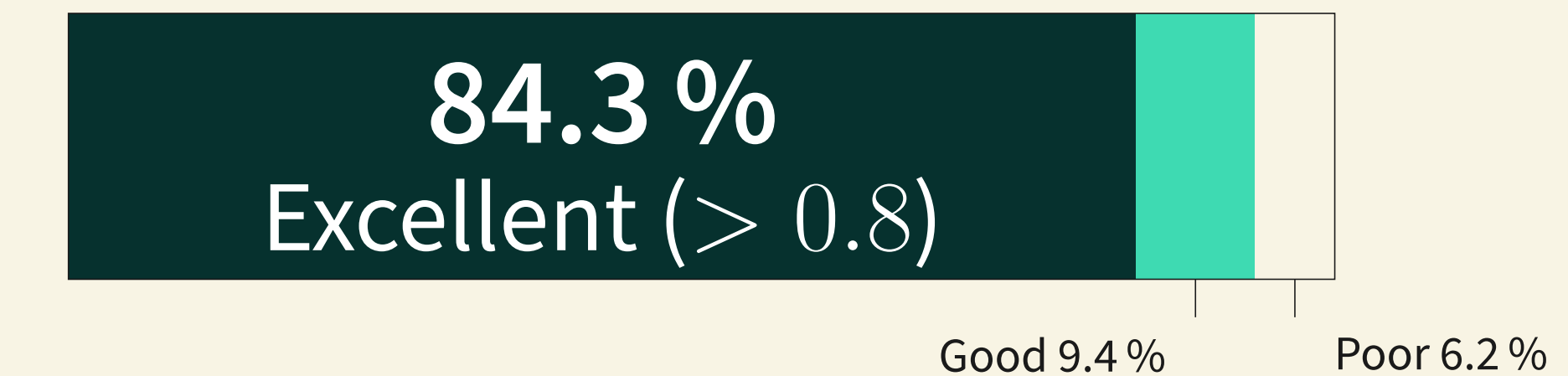


Figure 5. 84 % of feature labels are excellent under token-level fuzzing. 402 features, 4,422 A/B examples; combined score weights token-level 0.7 vs. prompt-level 0.3.

- **Combined score:** 0.912 ± 0.008 ($p < 10^{-4}$ vs. 0.5).
- **Token-level accuracy:** 0.906 ± 0.007 .
- Token-level evaluation catches labels right for the *wrong* reasons; prompt-level alone overestimates.

Causal Ablation

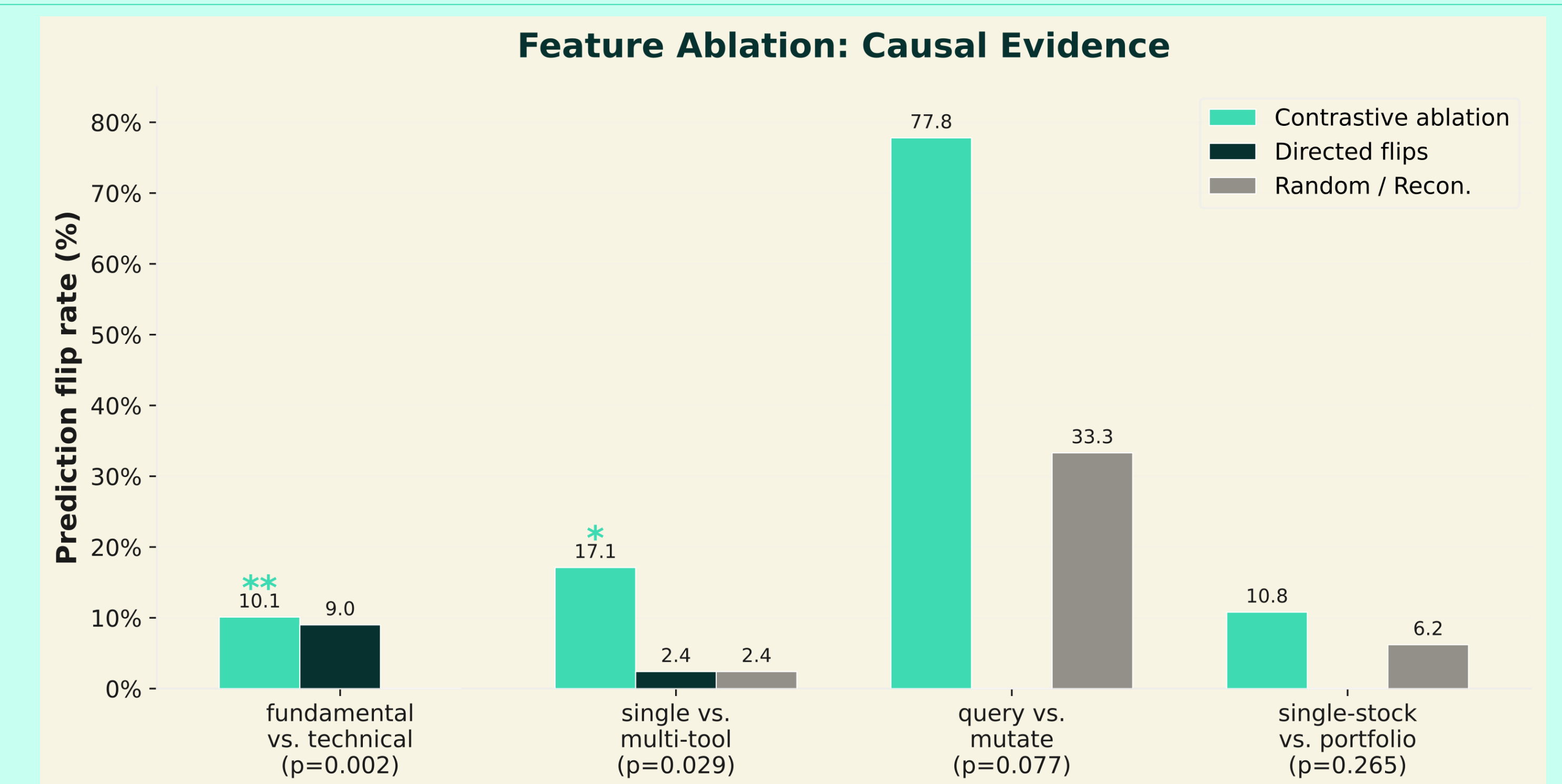


Figure 6. Ablating SAE features changes tool decisions — random features do not. Top-10 contrastive features vs. random features vs. SAE round-trip baseline.

Where it works (and where it doesn't)

- **Works:** *fundamental vs. technical* — 10.1 % flip, controls 0 % ($p = 0.002$).
- **Works:** *single vs. multi-tool* — 17.1 % vs. 2.4 % ($p = 0.029$).
- **Null:** 9 of 23 contrast types show zero flip rate. Those decisions are encoded *distributively* across many features, not localized in a sparse circuit. A known boundary — and a checklist item for replications.

Why the effect is real

Random-ablation flip rate *equals* SAE round-trip distortion across all 23 contrast types. The contrastive flip is **above** this noise floor — it must come from the specific features removed, not from encode/decode artefacts.

Code & Paper

Scan the QR (top-left) for the paper + code:

github.com/dataiku/kiji-inspector

Nemotron-3-Nano-30B (subject) · Qwen3-VL-235B (generation, labelling, judge) · Compute by NVIDIA

Prior work: [1] Bricken'23 · [2] Rajamanoharan'24 (JumpReLU) · [3] Juang'24 (autointerp).